

Investigating the Effect of Repeated Intermediate Data Encoding in Variational Quantum Circuits When the Number of Features Matches the Number of Available Qubits

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Abstract

Variational Quantum Circuits (VQCs) have emerged as promising architectures for near-term quantum machine learning due to their hybrid quantum-classical optimization framework and compatibility with Noisy Intermediate-Scale Quantum (NISQ) devices. However, a persistent challenge in VQC design is the mismatch between the limited number of available qubits and the high dimensionality of classical datasets. To address this limitation, data re-uploading, also referred to as intermediate data encoding, has been introduced as an approach in which classical features are repeatedly embedded between successive trainable variational layers rather than being encoded only once at the beginning of the circuit.

Existing studies primarily motivate intermediate encoding as a mechanism for overcoming qubit limitations by allowing high-dimensional classical information to be sequentially introduced into a restricted quantum register. However, a less explored question remains: does repeated intermediate encoding continue to provide benefits when the number of classical features already matches the number of available qubits? In this scenario, each feature can be directly assigned to an individual qubit during the initial encoding

stage, potentially removing the original motivation for repeated data injection.

This work investigates this question through a controlled experimental analysis of a Variational Quantum Circuit implemented using PennyLane. Four classical features from the UCI Student Performance dataset were mapped onto four qubits using an RX-RZ feature encoding strategy. The number of qubits, dataset, optimizer configuration, learning rate, batch size, and encoding method were kept constant, while the number of intermediate encoding stages was systematically varied from 1 to 12. The effect of increasing encoding depth was evaluated through training and testing accuracy measurements.

The results demonstrate that repeated intermediate encoding increases the expressive capacity of the VQC, as indicated by the continuous improvement of training performance with increasing circuit depth. However, testing performance does not improve monotonically and reaches its highest values at relatively shallow depths before declining as additional encoding stages are introduced. These observations suggest that while intermediate encoding enhances representational capacity even when qubit limitations are absent, excessive re-uploading increases model complexity without guaranteeing improved generalization. This study therefore highlights intermediate encoding as both a strategy for managing limited quantum resources and a mechanism for controlling the expressivity-generalization trade-off in variational quantum models.

1. Introduction

Variational Quantum Circuits (VQCs) represent one of the leading approaches for

hybrid quantum-classical learning on Noisy Intermediate-Scale Quantum (NISQ) devices. Their architecture typically consists of three fundamental components: classical data encoding, parameterized quantum transformations, and measurement. Among these components, the feature encoding mechanism plays a critical role, as it determines how classical information is embedded into the quantum state space and subsequently accessed by trainable quantum operations.

In conventional VQCs, classical features are encoded once at the beginning of the circuit. Following this initial embedding, subsequent variational layers transform the quantum state without direct reintroduction of the original classical information. Recent approaches based on data re-uploading, or intermediate data encoding, modify this structure by repeatedly injecting classical features between trainable layers. This enables each variational layer to interact with the original data representation rather than relying solely on previously transformed quantum states.

The primary motivation behind intermediate encoding has traditionally been the limited availability of qubits in current quantum hardware. Since the dimensionality of classical datasets often exceeds the number of accessible qubits, repeatedly encoding subsets of features provides a method for representing high-dimensional information within a constrained quantum system.

However, this raises a fundamental architectural question: when the number of classical features already matches the number of available qubits, does repeatedly encoding the same information across multiple variational layers still provide measurable advantages? In this setting, every feature can be directly assigned to a dedicated qubit during the initial encoding

stage, removing the immediate necessity for additional feature injection.

This work investigates this question through a controlled experimental study of intermediate encoding depth in a Variational Quantum Circuit. By maintaining the feature-to-qubit ratio at one-to-one correspondence while varying only the number of intermediate encoding stages, this study isolates the effect of repeated data encoding on model expressivity and generalization. The objective is to determine whether intermediate encoding provides benefits beyond its conventional role as a solution to limited quantum feature capacity.

2. Theoretical Framework & Mathematical Formulation

To rigorously analyze repeated intermediate data encoding, we formulate the hybrid circuit mathematically.

2.1 State Preparation and Layered Unitaries

Let $x = (x_1, x_2, x_3, x_4)^T \in R^4$ represent a normalized classical feature vector, and let θ_l denote the vector of trainable parameters for layer l .

In a traditional VQC, the quantum state is prepared by applying an encoding unitary $U(x)$ once to the initial state $|0\rangle^{\otimes n}$, followed by L variational layers $V(\theta_l)$:

$$|\psi(x, \theta)\rangle = \left(\prod_{l=1}^L V_l(\theta_l) \right) U(x) |0\rangle^{\otimes n}$$

In contrast, an interleaved data re-uploading VQC applies the encoding unitary $U(x)$ repeatedly prior to each variational block $V_l(\theta_l)$:

$$|\psi(x, \theta)\rangle = \left(\prod_{l=1}^L V_l(\theta_l) U_l(x) \right) |0\rangle^{\otimes n}$$

Where $U_l(x)$ utilizes an $RX - RZ$ product gate encoding on each qubit q :

$$U_l(x) = \bigotimes_{q=1}^n RZ(x_q) RX(x_q)$$

2.2 Measurement and Objective Function

The circuit output is derived from the expectation value of an observable operator M (e.g., $M = Z_0$):

$$\hat{y}(x; \theta) = \langle \psi(x, \theta) | M | \psi(x, \theta) \rangle$$

Model predictions $\hat{y} \in [-1, 1]$ are mapped to binary label probabilities, and the parameters θ are optimized using the Mean Squared Error (MSE) loss function over a dataset D :

$$L(\theta) = \frac{1}{|D|} \sum_{(x_i, y_i) \in D} \left(y_i - \hat{y}(x_i; \theta) \right)^2$$

3. Motivation and Research Gap

Existing literature predominantly considers data re-uploading as a strategy for addressing the limited availability of qubits in quantum computing systems. While this motivation remains highly relevant for high-dimensional datasets, it does not fully capture the potential architectural role of repeated feature encoding.

Each intermediate encoding stage introduces the classical features into a quantum state that has already undergone trainable transformations. Therefore, repeated encoding may allow subsequent variational layers to repeatedly interact with the original data representation, potentially increasing the complexity of the feature space and

enabling the circuit to learn richer nonlinear decision boundaries, even when additional qubits are not required.

This study isolates this architectural effect by maintaining a one-to-one correspondence between classical features and quantum resources, where the number of input features ($N=4$) exactly matches the number of available qubits ($n=4$). Under this controlled setting, any performance variation can be attributed primarily to changes in intermediate encoding depth rather than increased feature capacity.

4. Research Questions

The following research questions guide this investigation:

1. Does repeated intermediate encoding improve the performance of a Variational Quantum Circuit when the number of classical features already equals the number of available qubits?
2. How does increasing the number of intermediate encoding stages affect optimization performance and predictive accuracy across different circuit depths?
3. Does increasing encoding depth improve the model's ability to generalize to unseen data, or does it primarily increase the circuit's capacity to fit the training data?

5. Hypotheses

H1: Repeated intermediate encoding will increase the expressive capacity of the Variational Quantum Circuit, resulting in improved training performance as additional encoding stages provide more opportunities for interaction between classical features and trainable quantum transformations.

H2: Beyond an optimal circuit depth, additional intermediate encoding stages will

provide diminishing returns and may reduce generalization performance as increased circuit complexity allows the model to fit training-specific patterns more effectively. **H3:** An optimal encoding depth exists where the expressive capacity gained from additional intermediate encoding is balanced against the increased complexity introduced by deeper quantum circuits.

6. Experimental Design

6.1 Dataset Description and Feature Selection

The UCI Student Performance dataset contains 395 student records. To enable efficient repeated optimization across multiple circuit depths, a random subset of 200 samples was selected for experimentation. Four academic features (study time, failures, absences, and G1) were selected and mapped directly onto four qubits, creating a one-feature-per-qubit encoding regime ($N=n=4$). The target variable was converted into a binary classification task. Students with final grade $G3 \geq 10$ were assigned a positive label (+1), representing successful academic performance, while students with $G3 < 10$ were assigned a negative label (-1). The selected features were standardized using feature normalization and scaled to appropriate ranges for quantum rotation gate encoding.

6.2 Controlled Experimental Protocol

A controlled experimental protocol was adopted to isolate the effect of intermediate encoding depth. All hyperparameters were kept constant across experiments, with circuit depth serving as the only independent variable.

Fixed Hyperparameters:

- **Qubit Count:** 4 qubits ($n = 4$)
- **Dataset:** UCI Student Performance Dataset (4 selected academic features mapped directly to qubits)
- **Feature-to-Qubit Mapping:** One feature encoded per qubit ($N=n=4$)
- **Data Split:** $N_{train} = 80$ samples, $N_{test} = 20$ samples (80/20 train/test split)
- **Encoding Strategy:** $RX - RZ$ feature encoding
- **Optimizer:** Adam optimizer
- **Learning Rate & Batch Size:** Fixed across all configurations
- **Random Seed:** Fixed random initialization seed
- **Preprocessing:** Identical normalization pipeline

Independent Variable:

- **Circuit Depth (L):** 8 configurations evaluated (1, 2, 3, 4, 5, 6, 7, and 12 layers).

where each depth represents the number of repeated intermediate encoding and variational layer pairs.

7. Results

The performance metrics for each circuit depth configuration are detailed in *Table 1* below:

Layers (L)	Training Accuracy (%)	Testing Accuracy (%)
1	63.75%	70.00%

2	85.00%	65.00%
3	88.75%	65.00%
4	93.75%	70.00%
5	97.50%	45.00%
6	96.25%	60.00%
7	96.25%	55.00%
12	97.50%	55.00%

Table 1: Training and Testing Accuracy across Circuit Depths

A key experimental observation is that training accuracy increased almost monotonically as circuit depth increased, indicating that repeated intermediate encoding consistently enhanced the expressive capacity of the Variational Quantum Circuit. However, test accuracy exhibited a markedly different trend. Although shallow circuits (Layers 1 and 4) achieved the highest observed test accuracy of 70.00%, deeper architectures demonstrated reduced generalization performance, with accuracy declining to as low as 45.00% at Layer 5. This divergence between training and testing behavior suggests that increasing encoding depth improves the circuit's ability to fit the training data, but does not necessarily result in improved generalization to unseen samples.

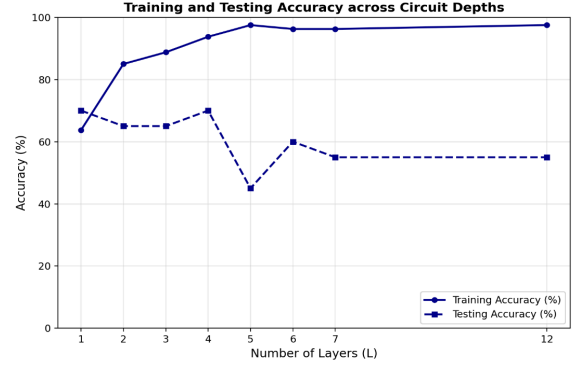


Figure 1: Training and testing accuracy as a function of intermediate encoding depth.

8. Discussion

The experimental results indicate that repeated intermediate encoding influences the Variational Quantum Circuit in two distinct ways.

First, increasing the number of encoding stages enhances the representational capacity of the quantum model. Each intermediate encoding stage reintroduces the classical input into a quantum state that has already undergone trainable transformations, allowing subsequent variational layers to repeatedly interact with both the original features and previously learned quantum representations. This increased interaction capability provides the circuit with greater flexibility for modeling complex nonlinear relationships, which is reflected in the continuous improvement of training accuracy, reaching 97.50% at the highest evaluated depth.

Second, the results demonstrate that increased expressive capacity does not necessarily translate into improved predictive performance. Although deeper circuits continued to optimize the training objective, their testing performance did not improve proportionally and instead showed a decline beyond certain depths. The

divergence between training and testing accuracy suggests that additional encoding stages increase the model's capacity to capture training-specific patterns, while providing limited benefit for learning patterns that generalize to unseen samples.

These observations are consistent with established principles in statistical learning theory, where increasing model capacity beyond an appropriate level can reduce generalization performance. In the context of Variational Quantum Circuits, intermediate encoding provides a mechanism for increasing effective circuit complexity without increasing the number of qubits. Therefore, while data re-uploading was initially introduced as a strategy for addressing qubit limitations, this study highlights an additional role: controlling the balance between representational capacity and generalization performance.

9. Relation to Existing Literature

The findings of this study are broadly consistent with *Is Data-Reuploading Really a Cheat Code? An Experimental Analysis*, which demonstrates that repeated data encoding can increase the expressive capability of Variational Quantum Circuits by allowing classical information to interact multiple times with evolving quantum states. The present experiments extend this observation by showing that the benefits of intermediate encoding can remain relevant even when qubit limitations are removed and each classical feature is assigned to an individual qubit.

However, this work examines a different aspect of data re-uploading from previous studies. Rather than focusing primarily on the ability of repeated encoding to represent

high-dimensional datasets using limited quantum resources, this study isolates the effect of encoding depth under a fixed feature-to-qubit correspondence. The observed divergence between training and testing performance suggests that while additional encoding stages continue to increase the circuit's expressive capacity, they do not guarantee improved generalization. These results emphasize that circuit depth should be selected carefully according to the desired balance between representation capability and predictive performance rather than assuming that deeper architectures will always provide superior results.

10. Conclusion & Future Work

This study investigated the effect of repeated intermediate data encoding in Variational Quantum Circuits under a controlled setting where the number of classical features matched the number of available qubits ($N=n=4$).

The experimental results demonstrate that increasing the number of intermediate encoding stages consistently improved the circuit's optimization capability, as reflected by increasing training accuracy. However, testing performance followed a different trend, reaching its highest observed accuracy in relatively shallow circuits (Layers 1 and 4, both achieving 70.00%) before declining as circuit depth increased. These results indicate that increasing encoding depth enhances the expressive capacity of the model, but additional representational power does not necessarily translate into improved generalization on unseen data.

Therefore, intermediate encoding should not be viewed solely as a strategy for overcoming qubit limitations. Instead, it should also be considered an architectural

parameter that controls the trade-off between quantum model expressivity, optimization capability, and generalization performance.

Code Availability

The implementation of the Variational Quantum Circuit, including data preprocessing, feature encoding, circuit construction, training pipeline, and experimental evaluation scripts, is publicly available at:

GitHub Repository: [VQC with Intermediate Encoding](#)

The repository contains all source code and configuration files required to reproduce the experiments described in this study.

analysis evaluates performance under a fixed initialization setting. Future iterations will perform multiple random seed initializations (e.g., 10 or more runs per circuit depth) and report mean accuracy with standard deviation to provide statistically robust performance comparisons.

Limitations & Future Directions:

1. **Dataset Geometry:** The experiments were conducted exclusively on a non-spherical educational dataset. Since the geometry of the data distribution may influence the effectiveness of quantum feature maps, future work will investigate whether similar behavior occurs on geometrically structured datasets, including spherical and synthetic benchmark datasets.
2. **Barren Plateaus vs. Overfitting:** Although the observed performance degradation is consistent with reduced generalization, further investigation is required to distinguish between capacity-related effects and optimization limitations. Future experiments will monitor gradient magnitudes across increasing circuit depths ($L \geq 12$) to evaluate whether barren plateau effects contribute to performance degradation.
3. **Statistical Reliability:** The current

References

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